

SMRG Flusim Web Dashboard

User Manual

This document will teach you how to use the SMRG Flusim Web Dashboard to run and visualise infectious disease simulations.

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1 Configuring Simulations

1.1 Setting Baseline Parameters

The SMRG Flusim Web Dashboard allows for many aspects of a simulation to be configured. In order to run a simulation, you must first set these model parameters to values matching the infectious disease you wish to simulate. Click on the “☰ Baseline Parameters” link on the sidebar (see [Figure 1](#)) to view the parameters that can be changed.

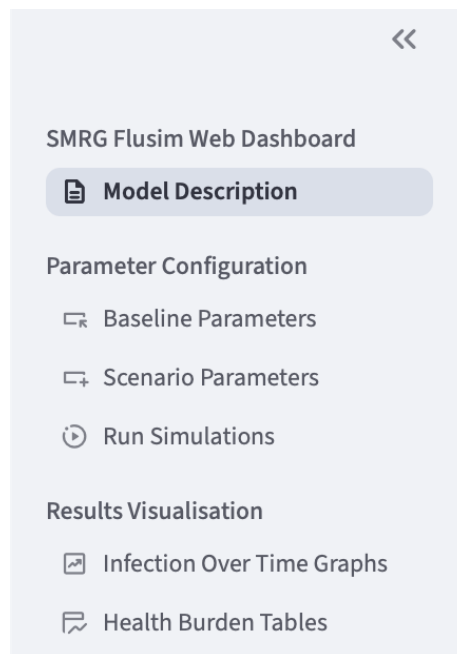


Figure 1: The navigation sidebar. Click on one of the page names to visit that page. If you can’t see this sidebar, click the » arrow on the top-left of the page to make it visible.

The parameters on this page are split into 4 tabs based on what aspect of the simulation they modify, as seen in [Figure 2](#). Click on any of these tabs to view its model parameters.

⚙️ Disease 👤 Community 🛡️ Vaccination and NPIs ⚙️ Dynamic

Disease Parameters

This tab contains parameters relating to the disease itself, including how it initially enters the community, the rate at which it spreads and how long infection lasts before recovery.

Note that despite being related to the disease's effects, hospitalisation and mortality rate are defined in the "Health Burden Outcomes" section of the "Community" tab instead of being in this tab, in order to group them with other health burden outcomes.

▼ Infection Seeding

These parameters control how infected individuals are directly seeded into the community. Seeding is typically used to kickstart the initial epidemic by ensuring a steady number of people are infected daily.

Infection Seeding Rate (Average Individuals per Day) ⓘ

0.25

Infection Seeding Time Period (Days) ⓘ

Day 1 Day 30

Figure 2: The ⚙️ Disease tab, with a pair of sliders controlling the rate at which infected individuals are seeded into the simulation model. These settings are used to initiate pathogen spread in the modelled community. Click on one of the tab names at the top to switch to that tab. Click and drag the circles on the sliders to change the infection seeding settings.

- ⚙️ **Disease** relates to the infection itself, including how frequently it infects new individuals and the length of time individuals will be infectious for.
- 👤 **Community** controls the simulated population and how they react to the disease, including the size of gatherings and how likely a sick individual is to stay at home.
- 💉 **Vaccinations and NPIs** controls vaccination strategies and [non-pharmaceutical interventions](#) (techniques used to limit the spread of disease without medication, e.g. social distancing).
- 🕒 **Dynamic** allows factors such as NPI effectiveness to change at specific times during the simulation period.

In each tab, several parameters may be adjusted. Most parameters use sliders for their input, as seen in [Figure 2](#). Click and drag the slider to change the value of its parameter. If you do not know what a specific parameter does, hover your mouse over the ⓘ symbol at the top-right of that parameter to show a description. Some parameters are enclosed in drop-down boxes like the one in [Figure 2](#); click on a box to open or close it.

1.2 Parameter Types

There are several different ways that parameters on the dashboard are changed. Below is a list of the different input methods.

Sliders

Sliders may be used to set time-based parameters on the dashboard. In addition to clicking and dragging the slider, you can also click a point on the bar to move the slider there immediately, or use the left and right arrow keys to fine-tune the values of sliders.

Some sliders, like the one in [Figure 3](#), have two values on the same line. The two values are the start and end of the range the parameter uses; either one can be adjusted freely.



Figure 3: A slider with two values to adjust. This selects a period of time over which infected people will be added to the simulation.

Toggles

Some parameters can either be enabled or disabled in the simulation. Parameters like the one shown in [Figure 4](#) can be toggled on and off by clicking the toggle switch. Note that some parameters are dependent on these toggles; for example, if “Enable Vaccines in Simulation” is toggled off you won’t be able to edit other vaccine-related parameters unless you toggle them back on.

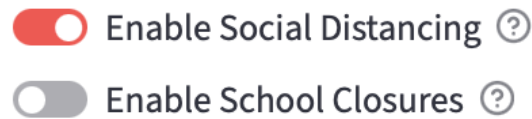


Figure 4: Toggle parameters which control whether specific [non-pharmaceutical interventions](#) are utilised in the simulation. The top one is enabled, while the bottom one is disabled.

Selections

If there are multiple options for a particular parameter, it will use a selection box like the one in [Figure 5](#). Click on the box to reveal the options, then click on the option you want. Remember that hovering your mouse over the ⓘ symbol will show a description of the parameter, including what each option relates to.

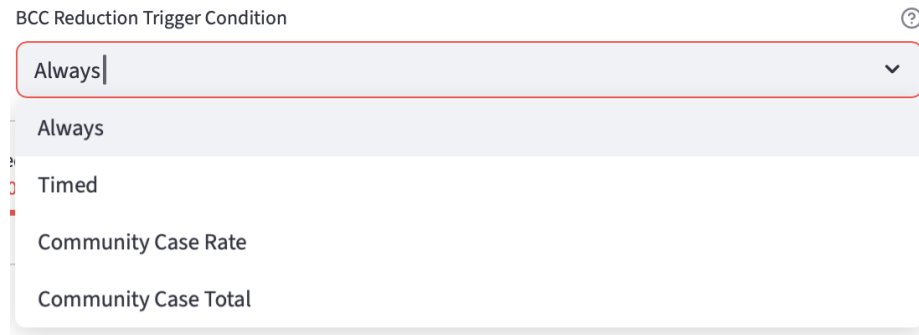


Figure 5: A parameter that uses a selection box to determine what type of condition will enable the [background contact count](#) reduction [NPI](#) in the simulation.

Number Inputs

Parameters like [Figure 6](#) require you to enter a number. Simply type the number you want and press Enter to confirm, or click the **+** and **-** symbols to increase or decrease the value by a small amount. If you enter an invalid value (e.g. a negative number), the dashboard will prompt you to enter something else.

First Dose Vaccination Rate (Vaccinations per Day) ?

- +

Figure 6: A number input parameter controlling how many people in the modelled community receive their first vaccination per day.

Tables

Some parameters can be set to a unique value for different age groups (see [Figure 7](#)). For parameters like these, a table is used to set the values for each age group. Double-click on a cell in the table to begin editing its value. For “Age Group” columns, select the age group that you would like to define age-specific parameter for using the drop-down menu. For numeric columns, simply type the new value and press Enter on your keyboard; if the column uses percentages, you will need to enter the value as a decimal.

Mortality Rate (Probability) ?

5%

Age-Specific Mortality Rate ?

Double-click a cell in this table to edit its value.

	≡ Age Group	≡ Mortality Rate
☐	Young Infant (0-6 Months)	20%
	Adult (25-44 Years)	7%

Figure 7: A parameter that uses a table to give different age groups different likelihoods of dying following infection.

You may add or remove rows from these tables freely. To add a row, hover your mouse over the row at the bottom of the table until a **+** icon appears, then click to add the row. Make sure to select an age group once the row has been added. Each row should use a different age group; if multiple rows have the same age group an error message will appear (see [Parameter Errors](#) for more details). To remove a row, click on the check box next to the row in the thin column to the left of the table, then click the  delete button in the top-right corner of the table. Note that these buttons will only be visible when your mouse cursor is hovering over their location.

The parameters in the  Dynamic tab (see [the descriptions of tabs in Setting Baseline Parameters](#)) also use tables for input. Instead of selecting an age group with the left column, you may enter a numbered day of the simulation where a parameter's value will change.

1.3 Additional Scenarios

To compare the effects that different parameter settings have on the simulation, the dashboard allows the definition of multiple parameter settings, or ‘[scenarios](#)’. Scenarios typically relate to alternative mitigation strategies involving vaccination and/or [non-pharmaceutical interventions](#), but can be used for any experiments that require comparing the spread of disease under different environments.

The parameters set on the “ Baseline Parameters” page are the values that will be used by the first scenario, also called the [baseline scenario](#). In order to add additional scenarios, use the sidebar to navigate to the “ Scenario Parameters” page. Click the “**+** Add Scenario” button to add an additional scenario to the dashboard.

On creating a new scenario, you will be presented with the panel seen in [Figure 8](#). Type a name in the “Name of Scenario” field and press Enter to give the scenario a name. This name will be used on all graphs and tables generated from the output of [simulation experiments](#) that include the scenario. The panel may also be used to configure the new scenario's parameters separately from the baseline scenario. By default, the new scenario's parameters will be set to the values chosen for the baseline scenario, but they may be changed freely. You may also remove the scenario by clicking the “ Remove Scenario” button below a scenario's name.

Scenario Parameter Configuration

Scenario #1

Name of Scenario ?

Scenario #1

Remove Scenario

Parameters

Disease Community Vaccination and NPIs Dynamic

Disease Parameters

This tab contains parameters relating to the disease itself, including how it initially enters the community, the rate at which it spreads and how long infection lasts before recovery.

Note that despite being related to the disease's effects, hospitalisation and mortality rate are defined in the "Health Burden Outcomes" section of the "Community" tab instead of being in this tab, in order to group them with other health burden outcomes.

> Infection Seeding

> Disease Transmission

> Disease Life Cycle

> Immunity Waning

+ Add Scenario

Figure 8: The panel where additional modelling [scenarios](#) may be created. Enter a name in the "Name of Scenario" field, and change parameters from their [baseline](#) values using the parameter tabs. Click "+ Add Scenario" to add an additional scenario to the simulation.

2 Running Simulations

To run [scenarios](#) as part of a [simulation experiment](#), use the sidebar to navigate to the “🕒 Run Simulations” page. This page (depicted in [Figure 9](#)) contains several parameters which control the settings of the [simulation engine](#). These settings control aspects like how long the simulation will run for (in days), how many [simulation runs](#) to perform with each scenario (to account for [stochasticity](#)) and what community the model will be based on. Simulation engine settings are applied to all scenarios rather than being individually configurable.

Simulation Engine Settings

Simulated Community ⓘ
Newcastle ▼

Length of Simulation (Days) ⓘ
360 Days

Number of Simulation Runs ⓘ
24

Simulation Starting Day of the Week ⓘ
Monday

🕒 Run Simulations

Figure 9: The “🕒 Run Simulations” page, with settings to configure the [simulation engine](#). Click the “🕒 Run Simulations” button below them to run your [scenarios](#) together.

Once the simulation engine settings are configured, click the “🕒 Run Simulations” button. The confirmation prompt seen in [Figure 10](#) will appear on the screen, listing the names of all configured scenarios. If there are any errors that result from the chosen parameter settings, such as an [NPI](#) being set to begin after the simulation has already ended, they will be listed beneath the parameter names (see [Parameter Errors](#) for more details). Click “Confirm” at the bottom of the prompt to run the simulation experiment.

When you run a simulation experiment, the dashboard sends the parameters for your scenarios to a separate server. This server will use these parameters to run the infectious disease model using the simulation engine. Due to the use of [stochastic](#) parameters (such as the probability of transmission between two individuals), a single simulation experiment will involve multiple runs of the simulation engine for each scenario. Once all simulations are complete, the server will use the outcomes of each simulation run to calculate the median daily infection counts for each scenario, then send these results back to the user dashboard. While this process is occurring, you will not be able to run any new scenarios, but can still navigate to other pages on the dashboard.

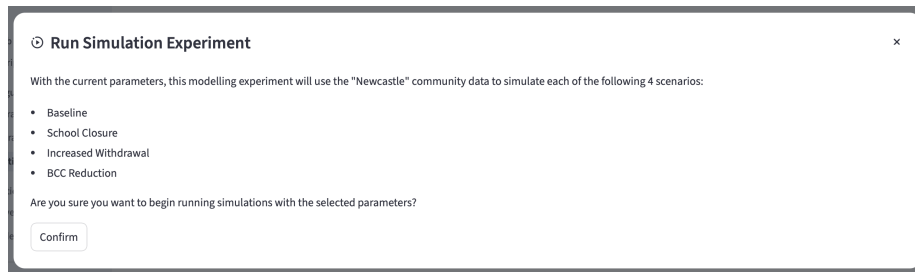


Figure 10: An example of the confirmation prompt that will appear when you attempt to run the simulation. In this instance, the user has added 3 intervention [scenarios](#), naming them “School Closure”, “Increased Withdrawal” and “[BCC](#) Reduction”. Click “Confirm” to send the parameters to the [simulation engine](#).

Each simulation experiment may take up to 30 minutes to run depending on how many scenarios were created. When the simulation is complete, a popup like the one in [Figure 11](#) will appear in the top-right corner of the webpage to inform you that the results are ready to be viewed.

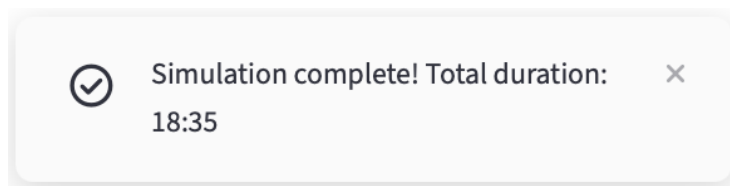


Figure 11: The popup that appears when the simulation is complete, which also lists how long it took to run.

3 Visualising Simulation Results

Once a [simulation experiment](#) has finished running and the server has returned its results, you will be able to use these results to generate graphs and tables.

3.1 Infection-Over-Time Graphs

The simplest of the visualisations the dashboard can create is a line graph plotting median infections over time (averaged over the multiple [simulation runs](#) in each [experiment](#)), such as the classical epidemic curves displayed in [Figure 12](#). Use the sidebar to navigate to the “[Infection Over Time Graphs](#)” page in order to generate these graphs.

Median Infections per Day Over Time

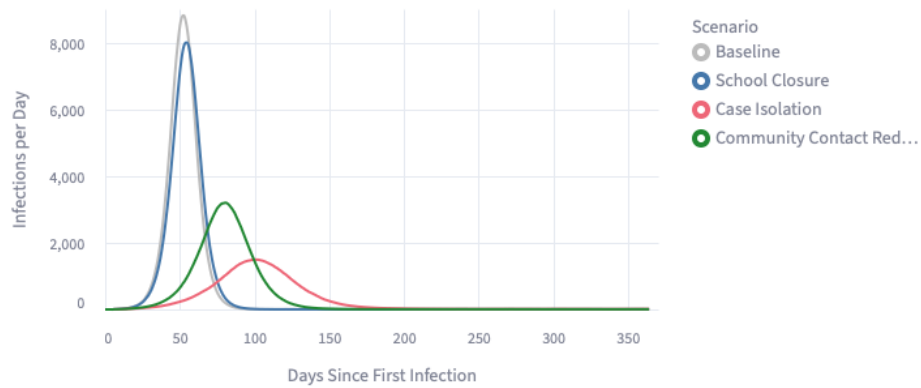


Figure 12: An example of an infection-over-time graph that can be generated using the dashboard, plotting daily infection curves under alternative mitigation measures.

Clicking on the “Graph Settings” menu seen in [Figure 13](#) lets you configure whether the graph should display daily infections or total (cumulative) infections. It also contains a multiple selection box controlling which [scenarios](#) are displayed on the graph; by default, all scenarios defined in the “[Scenario Parameters](#)” page are included. Click the **×** next to a scenario to remove it, or click the **⊗** on the right to remove all of them; click the box itself to display any scenarios that can be added.

Click the “[Create Graph](#)” button to generate the graph. If the “[Create Graph](#)” button is greyed out, it may mean that your simulation hasn’t finished running yet; wait until it is complete before trying to generate a graph.

Figure 13: The settings panel for infection-over-time graphs, allowing you to choose the type of graph and which [scenarios](#) to include. Click the “ Create Graph” button to use these settings to generate a graph.

Infection-over-time graphs are displayed beneath the “Graph Settings” menu. To download a copy of the data used to generate the graph, click the “ Download Infection Data” button. Further options for viewing the graph can be seen in [Appendix A](#) under [Infection-Over-Time Graph Configuration](#).

3.2 Health Burden Tables

The dashboard may be used to create health outcome tables from the [simulation experiment](#)’s generated data, compiling [health burden outcomes](#) as in [Figure 15](#). These tables display the resulting data as metrics such as diagnosed cases, hospitalisations and deaths. The figures for these tables are made by combining the age-specific infection data extracted from the simulation output with probabilities defining the occurrence of health burden outcomes relative to infections. These probabilities are defined in the Disease tab when configuring the parameter settings for the experiment. Navigate to the “ Health Burden Tables” page using the sidebar to create these tables.

Tables use a more advanced settings menu than graphs, as in [Figure 14](#).

- [Scenarios](#) may be added or removed from the table, just as with infection-over-time graphs. Click the **x** beside each scenario to remove it, or click the on the right to remove all of them. Click the selection box itself to display all the scenarios that can be added.

The columns that will be included in the table can be configured individually using the table seen in [Figure 16](#), which works similarly to the tables used for age-specific parameters (see [Tables](#)). Double-click on a cell to begin editing it. Each row in the settings table represents a column that will be included in the final health burden table; rows can be added by clicking on the **+** icons on the bottom row of the table, or removed by clicking the ☐ checkbox on the left side of the table then the  delete icon at the top-right. Note that you may need to hover your mouse over the table to see these icons.

Select Health Burden Columns

Double-click a cell in this table to edit its value.







 Health Burden Outcome	 Options
Infections	
Deaths	Difference from Baseline
Deaths	Percentage Difference from Baseline
+	

Figure 16: The column selection form for health burden tables. 3 columns have been configured: a column displaying total infections for each [scenario](#), a column displaying the difference in total deaths for each scenario compared to the [baseline scenario](#), and a column displaying the percentage difference in deaths from the baseline for each scenario.

The left column lets you select which health burden the column should display. These health burdens are calculated using the rates defined under the “Health Burden Outcomes” section of the  Disease parameters (see [Configuring Simulations](#)); this allows different scenarios to model diseases with different levels of severity. The right column lets you select options that modify how the columns are formatted. This column allows for multiple selections per row; you may select as many options as you like for each column, or none at all. The options that are available for each column are as follows:

- The “Difference from Baseline” option makes the column display the difference in health burdens between a given scenario and the [baseline scenario](#), rather than displaying the raw values.
- The “Percentage” option makes the column display the percentage of the population instead of the raw values.
- If both “Difference from Baseline” and “Percentage” are selected, the column will display the percentage difference in health burdens between a given scenario and the baseline scenario.

Once you have selected your chosen settings, click the “ Create Table” button to generate the table. Like the “ Create Graph” button, the “ Create Table” button may be greyed out if you have yet to complete a simulation or if no columns have been configured.

To download a copy of the generated table, click the “ Download Table Data” button. Further options for viewing the table can be seen in [Appendix A](#) under [Table Configuration](#).

4 Additional Notes

This section contains notes on what to do if errors occur alongside other features of the dashboard.

4.1 Appearance Settings

Clicking the  menu dots at the top-right corner of the screen and selecting “Settings” from the options that appear will allow you to modify the dashboard’s appearance. From here, you may toggle wide mode (stretching the contents of the dashboard to reach the edge of the screen) or switch between light and dark website themes.

4.2 Simulation Errors

If the dashboard cannot connect to the server, you will see the popup in [Figure 17](#) appear when attempting to run the simulation. Ensure you are connected to the internet and try again. Other issues with the simulation will cause similar popups to appear; the error message will explain what to do.

4.3 Parameter Errors

If you set a parameter to a value that would cause issues in the simulation, an error message will appear above that parameter. The error message will explain why the current values cause issues, and give you multiple options for how you can fix the issue. The error message will also appear at the top of the parameter selection page to ensure it is not overlooked.

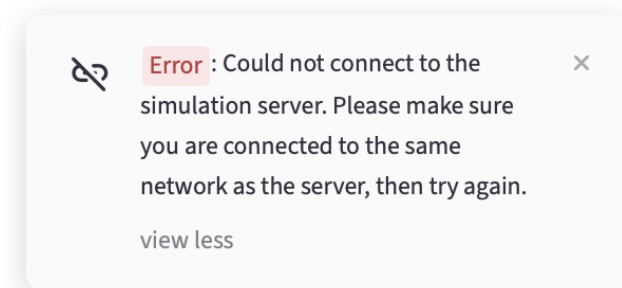


Figure 17: The popup that appears if the dashboard cannot connect to the server and run the simulation.

Error messages may be either yellow or red. Yellow error messages are warnings; they appear if parameters will cause unrealistic or unintuitive behaviour. You may still run a simulation if your parameters receive yellow error messages, but you should read them carefully as they are likely to result from an oversight when setting parameters. Red error messages are crucial errors; the simulation will not run if any of these are present. [Figure 18](#) shows examples of both red and yellow errors for the same parameter.

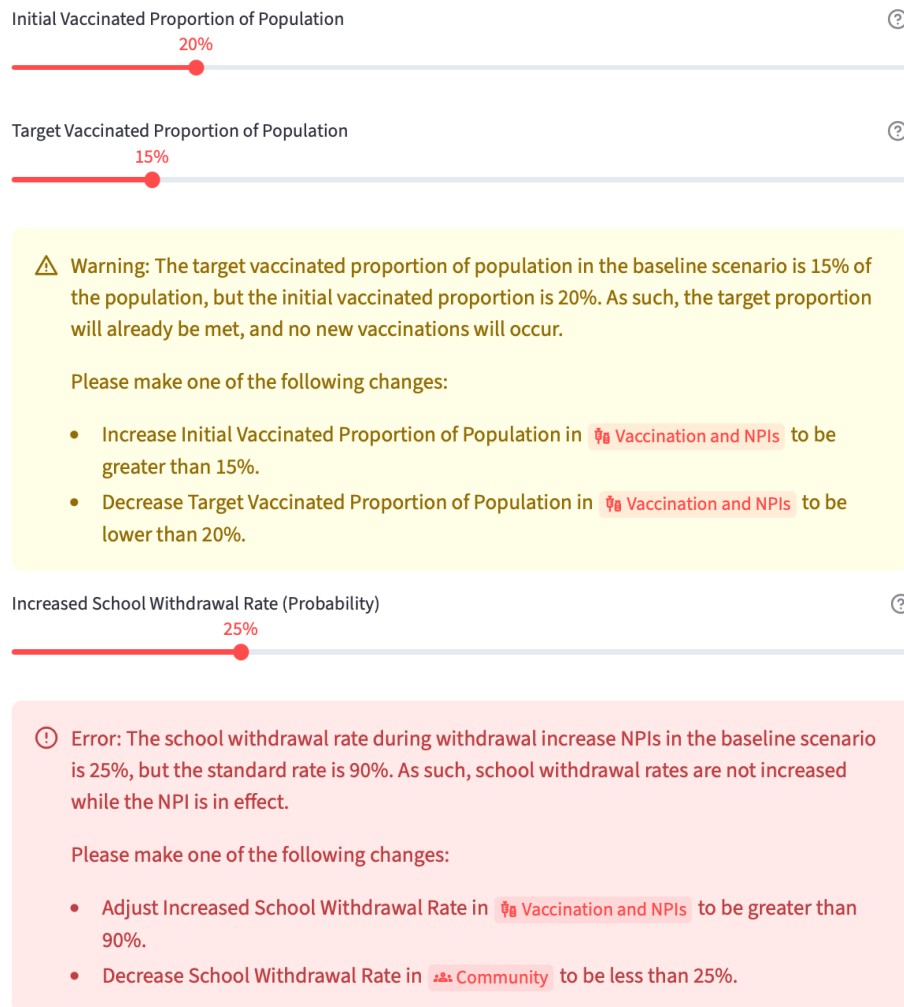


Figure 18: The two different types of error message that can appear next to parameters. Yellow messages are warnings, while red messages are crucial errors.

Glossary

***Flusim* model** The agent-based simulation model used by this dashboard to simulate infectious respiratory diseases. [18](#)

background contact count A parameter defined as part of the [Flusim model](#), which determines how many interactions between random people in the simulation occur each day. It is used to emulate interactions that occur outside of the model’s simulated locations, such as on public transport or in sporting events. [5](#), [10](#)

baseline scenario The [scenario](#) defined in the “ Baseline Parameters” page. All other scenarios within a [simulation experiment](#) will use the parameters defined in this scenario as default values for parameters that the scenarios do not have their own values for. [7](#), [8](#), [14](#)

health burden outcome A statistic measuring a negative consequence of an infectious disease outbreak. Examples include GP visits, hospitalisations and deaths. [12](#), [18](#)

non-pharmaceutical intervention Any intervention or protocol implemented to combat the spread of disease that does not involve medication or pharmaceutical products. Examples include social distancing and school closure. [4](#), [5](#), [7](#), [9](#)

scenario A set of parameters defining a single environment for the simulation. [7–14](#), [18](#)

simulation engine The program that conducts simulations according to the [Flusim model](#), hosted separately from the dashboard itself. [9](#), [10](#), [18](#)

simulation experiment The process that is triggered upon clicking the “ Run Simulations” button, in which multiple [scenarios](#) are ran together by the [simulation engine](#) in order to compare infection rates and [health burden outcomes](#) between them. [7](#), [9](#), [11](#), [12](#), [18](#)

simulation run The act of using the [simulation engine](#) to simulate a given [scenario](#) a single time. [Simulation experiments](#) use multiple runs for each scenario to account for the presence of [stochastic](#) parameters. [9](#), [11](#)

stochasticity The property of being random or unpredictable. Several parameters used by the [Flusim model](#) are stochastic in order to replicate the unpredictable nature of infectious diseases. [9](#), [18](#)

A Advanced Visualisation Options

As part of the Streamlit framework used to construct the dashboard, the graphs and tables throughout the dashboard have many interactive options to change how they are displayed. This appendix will list all of these options.

A.1 Table Configuration

Tables, like the ones used to configure age-specific parameters (see [Tables](#)) or the health burden tables that can be generated using the results of simulation experiments (see [Health Burden Tables](#)) utilise many forms of interactivity. Note that many of the buttons and icons mentioned here are not visible unless your mouse is hovered over the table.

- If the generated table is too large for the dashboard to display at once, you can scroll the table both vertically and horizontally using the scroll bars that appear on the right and bottom edges. The scroll wheel on an optical mouse will also work; holding Shift will allow you to scroll horizontally with the scroll wheel instead of vertically.
- The Scenario and Age Group columns in health burden tables use unique colours for each scenario/age group in the table. Additionally, any “Difference from Baseline” columns are colour-coded based on the size of the difference; greater decreases in health burden outcomes are more blue, while greater increases are more red.
- Clicking on a cell will highlight it. Clicking and dragging allows multiple cells to be highlighted. When cells are highlighted, the standard copy hotkeys (`ctrl` + `C`, or `⌘` + `C` on Mac computers) can be used to copy the values in the highlighted cells to your clipboard.
- Double-clicking on a cell containing a number will display the exact value of that cell, without any rounding or other formatting.
- Hovering your mouse over a column header will show a tooltip explaining what the values in the column represent.
- Clicking on a column header will sort the table using that column, in ascending order. Clicking again will switch to descending order, and clicking a third time will remove the sorting. Note that only one column can be used for sorting at a time; clicking on another column header will remove the current sorting.
- Clicking and dragging a column header left or right allows you to reorder the columns in the table.

- Clicking and dragging the borders between column headers allows you to adjust how wide each column is.
- Clicking on the  menu dots to the right of a column header will open a drop-down menu where different settings for the column can be modified, seen in [Figure A1](#). From here, the following options are available:

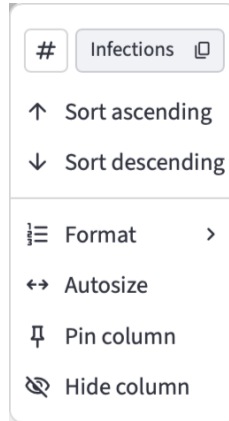


Figure A1: An example of the list of options that can be used to configure a column's appearance within a table.

- Clicking the  copy symbol next to the column's name will copy the name of the column to the clipboard.
- The  Sort ascending and  Sort descending options will sort the table in the same way as clicking on the header does.
- The  Format option on numeric columns allows you to change how the column's numbers are formatted. The available options include  Plain (no formatting or rounding),  Compact (uses suffixes like K for 1000 to abbreviate numbers),  Percent (displays values as percentages rounded to 2 decimal places) and  Scientific (displays values using scientific notation, e.g. 1.873e5), among other, more specialised options.
- The  Autosize option will adjust the width of the column so that it is exactly enough to display the column header and all currently-visible cells without cutting anything off.
- The  Pin column option will force the column to always appear on the left edge of the table, even if the table is horizontally scrolled to the right. The Scenario and Age Group columns in health burden tables are pinned by default; the options for these columns will include an  Unpin column option to disable this.
- The  Hide column option will remove the column from the table.

- The top-right corner of a table includes settings to configure the table as a whole. The following options can be selected here:
 - The  eye icon opens a menu where columns can be added or removed from the table. Columns hidden from their own menu can be re-added here; click the checkboxes next to each column name to show or hide the column. Most tables also have an index column that is hidden by default, but can be added through this menu.
 - The  download icon will download a CSV copy of the table. Note that sorting, removing or reordering columns will affect the CSV version of the table as well.
 - The  search icon will open up a panel allowing you to search for specific values. Typing a specific value and pressing Enter will highlight cells that contain that value. From here, you may use the   arrow icons to jump between these cells.
 - The  fullscreen icon will expand the table to fill the entire webpage. To return the table to normal size, use the  exit fullscreen icon.
 - In tables used to configure settings, the  add icon will add a blank row to the bottom of the table. If any rows have been selected using the checkboxes on the left edge of the table, this icon will change to a  delete icon, which will remove the selected rows from the table.

A.2 Infection-Over-Time Graph Configuration

Like tables, infection-over-time graphs incorporate several interactive features.

- Hovering your mouse over the graph will display a tooltip showing the values for each scenario at that time-point on the graph.
- Clicking on the name of a scenario to the right of the graph will hide the lines for all scenarios except that one. Hold Shift while clicking on another scenario name to make that scenario's line reappear without hiding other visible scenarios.
- Clicking the  fullscreen icon in the top-right corner will expand the image to fill the entire webpage. To return the image to normal size, click the  exit fullscreen icon.
- Clicking the  chart icon in the top-right corner will hide the graph and display a spreadsheet containing the data used to generate the graph. This spreadsheet uses many of the same options for formatting that other tables on the dashboard use (see [Table Configuration](#)). To return to displaying the graph, click the  graph icon.
- Clicking the  menu dots will display a list of miscellaneous options. The Save as SVG and Save as PNG options allow the graph to be downloaded as an image in either file format. The other options allow you to view the Vega code that is used to generate the graph.